

Saurabh Kumar

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EDUCATION

Dr. APJ Abdul Kalam Technical University
Bachelor of Technology in Computer Science Engineering
• CGPA: 7.39/10.0

Lucknow, Uttar Pradesh
Nov. 2023 – June 2027

Uttar Pradesh Board (UPMSP)
Junior Secondary (Class X)
• Percentage: 87%

Uttar Pradesh
2020

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), Python, C++, SQL, HTML5, CSS3
Frontend Development: React.js, Redux, Responsive Web Design, Single Page Applications (SPA), DOM Manipulation, Bootstrap, UI/UX Design
Backend Development: Node.js, Express.js, RESTful API Design, API Integration, Microservices Architecture, Server-Side Rendering, Real-time Systems
Database Management: MongoDB, Mongoose ODM, MySQL, Database Design, Query Optimization, Data Modeling, NoSQL, Relational Databases
Machine Learning & AI: TensorFlow, Keras, Deep Learning, Neural Networks, CNN, Computer Vision, Image Processing, NLP (Natural Language Processing), LLM (Large Language Models), Google Gemini API, Model Training, Transfer Learning
DevOps & Tools: Git, GitHub, Version Control, Docker (Containerization), Linux (Ubuntu), Postman, VS Code, Jupyter Notebook, npm, Vercel, Render, Cloud Deployment
Software Engineering: Agile Methodology, Object-Oriented Programming (OOP), Data Structures, Algorithms, Problem Solving, Code Review, Full-Stack Development
Security & Authentication: JWT, Session Management, OAuth, OTP Verification, DDoS Protection, Rate Limiting, HTTPS, CORS

PROJECTS

- StudyMate - AI-Powered Study Platform** | *React, Node.js, MongoDB, Express, Gemini AI* 2024
- Engineered intelligent study companion leveraging Google Gemini AI with features including conversational AI tutor, automated document summarization, dynamic quiz generation, and comprehensive analytics dashboard with streak tracking.
 - Architected secure authentication system utilizing OTP verification, JWT-based session management, and implemented DDoS mitigation through rate limiting middleware, serving 20+ concurrent users.
 - Constructed scalable RESTful API architecture using Express.js and MongoDB with optimized query performance for real-time AI response generation and data persistence.
 - Crafted responsive single-page application with React.js following modern UI/UX design patterns and seamlessly integrated Google Gemini API for personalized learning recommendations.
 - GitHub:** github.com/saurabh-kumar135/StudyMate
- Brain Tumor Detection System** | *React, Node.js, Python, TensorFlow, Deep Learning* 2024
- Developed medical diagnostic AI system utilizing Convolutional Neural Networks (CNN) with TensorFlow and Keras for automated brain tumor detection from MRI scans.
 - Achieved superior accuracy on medical imaging dataset through hyperparameter tuning, data augmentation, and transfer learning techniques.
 - Delivered end-to-end full-stack solution featuring hospital-specific authentication, secure MRI upload interface, and real-time prediction engine with confidence scoring.
 - Orchestrated seamless integration between Python-based machine learning model and Node.js backend API using microservices architecture for production deployment.
 - GitHub:** github.com/saurabh-kumar135/brain-tumor-detection

HavenTo - Accommodation Booking Platform | *React, Node.js, MongoDB, Express* 2024

- Launched production-ready accommodation marketplace featuring property listings, user authentication, favorites management, and booking workflow similar to Airbnb.
- Deployed cloud-native architecture with frontend hosted on Vercel and backend microservices on Render, ensuring 99.9% uptime and scalability.
- Architected comprehensive RESTful API suite for property CRUD operations, JWT-based authentication, and stateful session management.
- Established responsive React.js interface with reusable component library and optimized MongoDB schema design for high-performance data retrieval.
- **GitHub:** github.com/saurabh-kumar135/HavenTo | **Live:** havento.vercel.app

ACHIEVEMENTS

Open Source Contributions: Actively contributing to open-source projects on GitHub with over 20 repositories.

Problem Solving: Solved over 100 coding problems across various platforms including LeetCode and HackerRank.

Project Portfolio: Successfully built and deployed three production-ready full-stack applications with modern tech stacks.

Technical Writing: Documented all projects with comprehensive README files and technical documentation.

AREAS OF INTEREST

Full Stack Web Development, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Artificial Intelligence, Cloud Computing, Quantum Computing